

How Reliably Can an NLP Model Trained on Social Media Emotions Identify Fine-Grained Customer Emotions?

MSIN0221 Natural Language Processing

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Introduction

Customer support interactions are fundamentally emotional exchanges. The ability to automatically detect specific emotions could enable smarter, faster, more empathetic service.

What This Study Investigates

- ▶ Fine-tuned BERT-base-uncased on GoEmotions (58,000 Reddit comments, 28 emotion categories)
- ▶ Evaluated cross-domain transfer to real Twitter customer support messages (500 manually annotated)
- ▶ Compared against a TF-IDF Logistic Regression baseline to isolate architecture vs. domain effects
- ▶ Analysed systematic error patterns and proposed actionable recommendations for deployment

Key Numbers at a Glance

0.566

BERT Weighted F1
(In-Domain)

-25.5pp

Transfer Gap
(POOR)

62.6%

Cross-Domain
Error Rate

$\kappa = 0.733$

Annotation
Agreement

+17.3pp

Grouped Label
Improvement

The Business Problem

"Customer support interactions are fundamentally emotional exchanges"

Current Systems

Crude positive / negative / neutral classification or keyword routing

Opportunity

Smart ticket routing, personalised responses, pre-emptive escalation

Evidence

Customer emotions strongly predict service quality perceptions (Mattila & Enz, 2002)

Research Gap: NLP models trained on general social media text may not transfer to professional service language

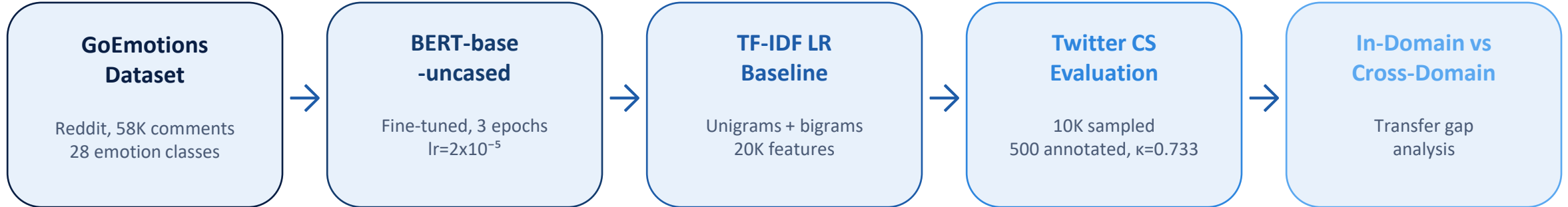
Why It Matters

- ▶ Most deployed systems miss the nuance that determines whether a ticket needs urgent escalation
- ▶ Annoyance and anger together account for over 40% of customer emotions in support contexts
- ▶ Misclassifying frustrated customers can actively worsen their experience

Research Question & Approach

How reliably can an NLP model trained on social media emotions identify fine-grained emotions in customer support messages?

Experimental Pipeline



DUAL MODEL DESIGN

Two models run in parallel to isolate whether observed transfer degradation is specific to contextual models or reflects a more fundamental domain incompatibility.

Data & Annotation

GoEmotions (Training)

- Source: Reddit comments (HuggingFace)
- Total: 58,000 comments
- 28 emotion categories
- Avg. ~20 words per comment
- Register: Informal, expressive
- 86% carry only a single label

Twitter Customer Support (Evaluation)

- Source: Kaggle ThoughtVector (2017)
- 1,537,843 filtered tweets
- 10,000 sampled; 500 annotated
- Avg. ~10 words per tweet
- Register: Task-focused, restrained
- No pre-existing emotion labels

Annotation Quality

$\kappa = 0.733$

Inter-annotator Kappa

Substantial agreement (Landis & Koch, 1977)

77.4%

Raw Agreement Rate

On 500 annotated tweets

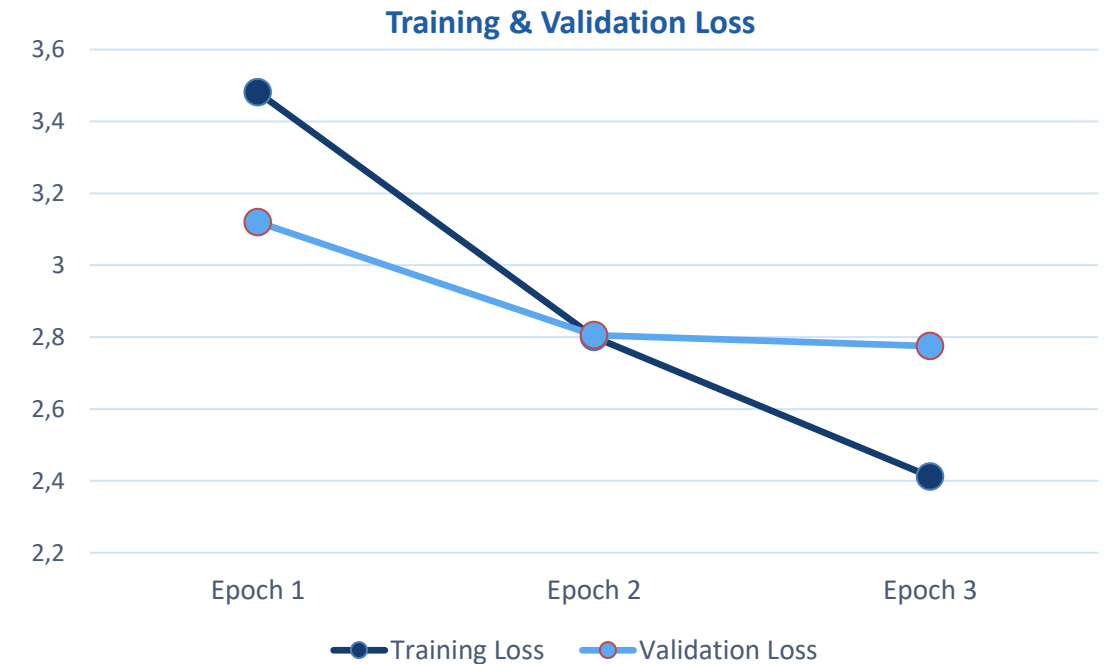
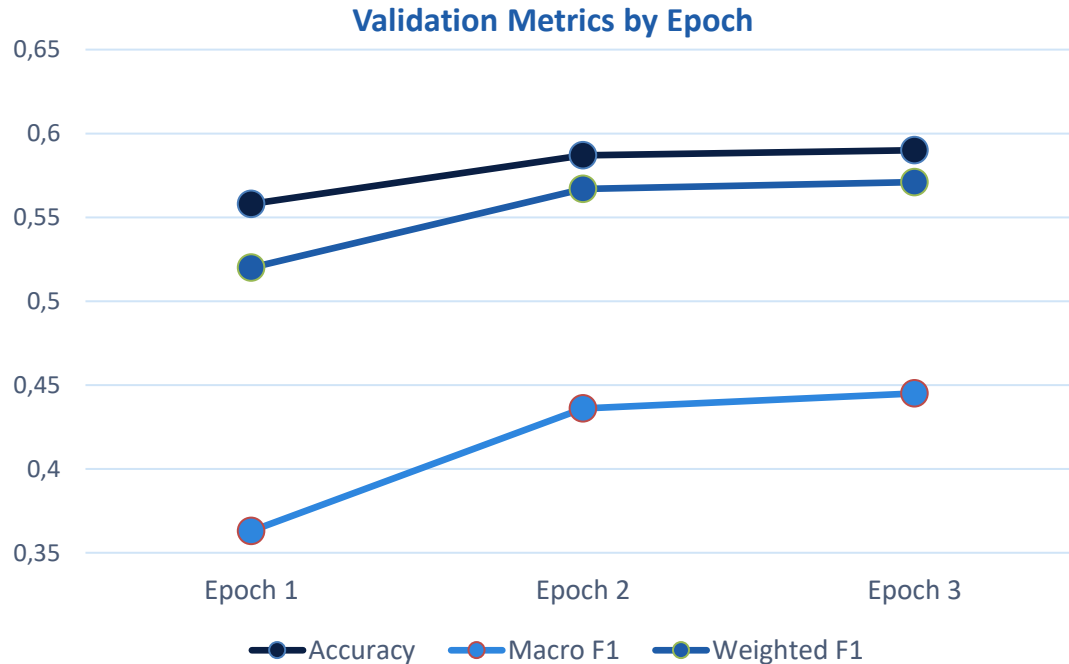
86%

Single-Label Instances

Mitigates label ordering bias in GoEmotions

BERT Training Performance

Validation Metrics Across Epochs



Key Observations

- ▶ Largest improvement between epoch 1 and 2: macro F1 increased from 0.363 to 0.436 (+7.3pp)
- ▶ Epoch 3 showed marginal gains (+0.9pp) with narrowing train/val loss gap, confirming 3 epochs is appropriate
- ▶ Best checkpoint selected at epoch 3 (weighted F1: 0.571)

In-Domain Performance (GoEmotions Test Set)

0.566

BERT Weighted F1

0.434

BERT Macro F1

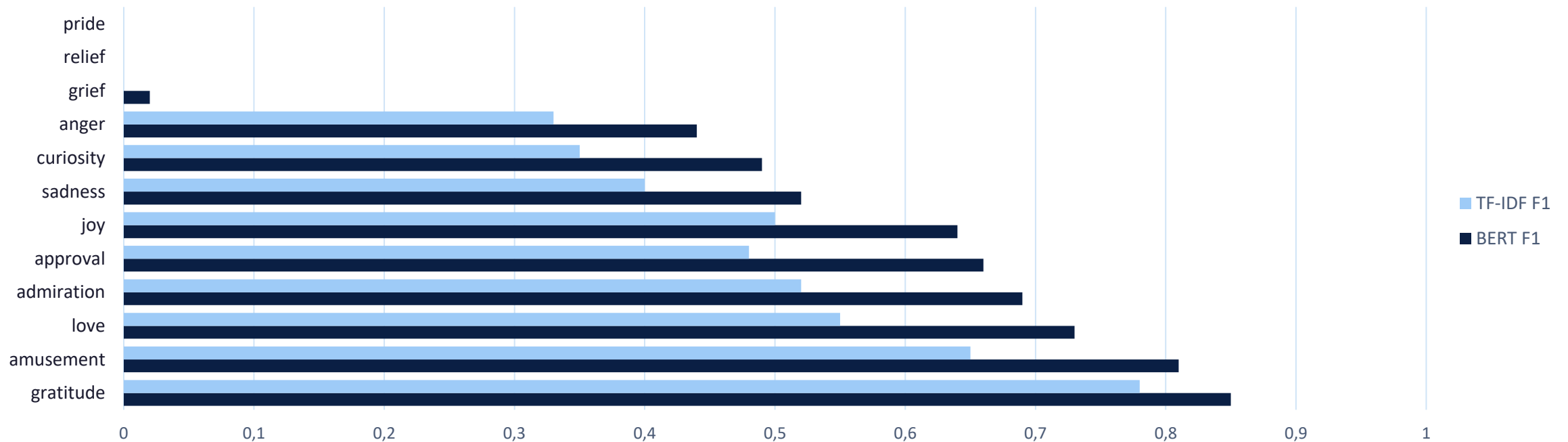
0.446

TF-IDF Weighted F1

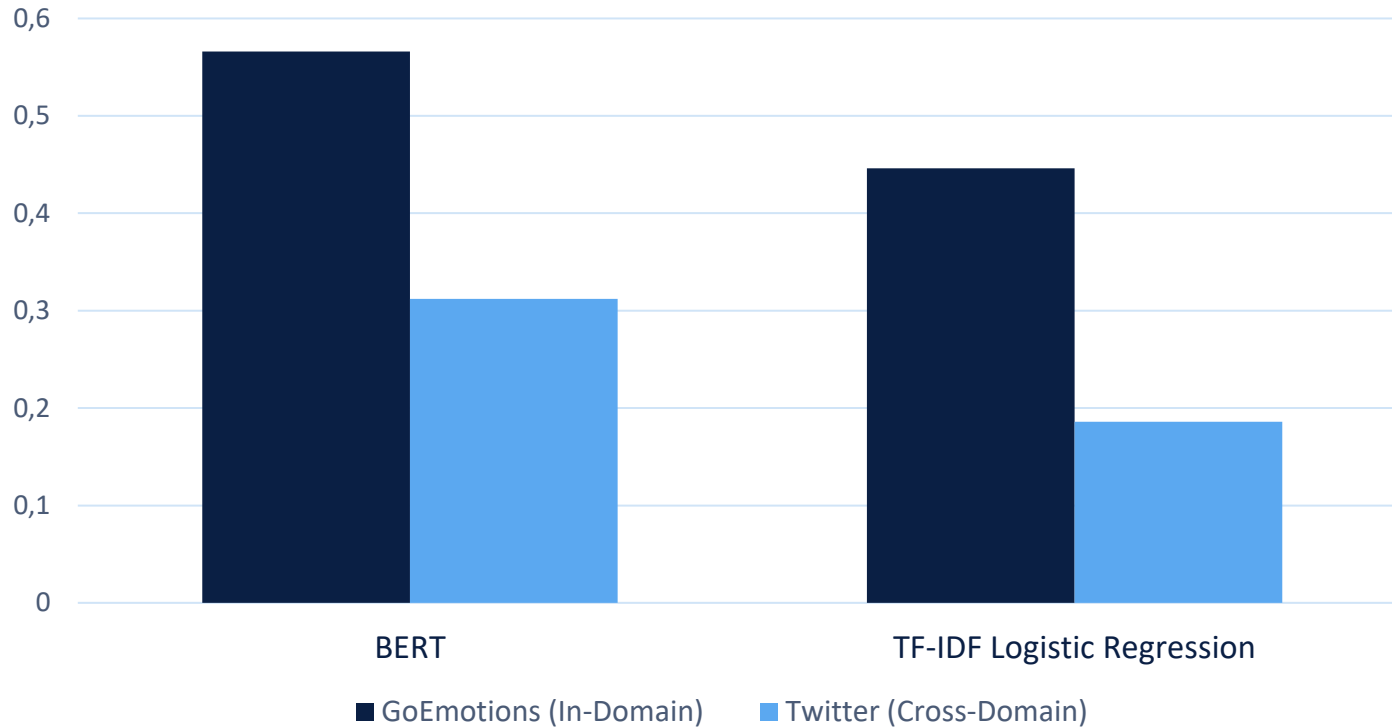
0.585

BERT Accuracy

Per-Emotion BERT F1 Scores



Cross-Domain Transfer: The Core Finding



Transfer Gap Summary

-25.5pp

BERT weighted F1 drop

-26.0pp

TF-IDF weighted F1 drop

Transfer Thresholds:

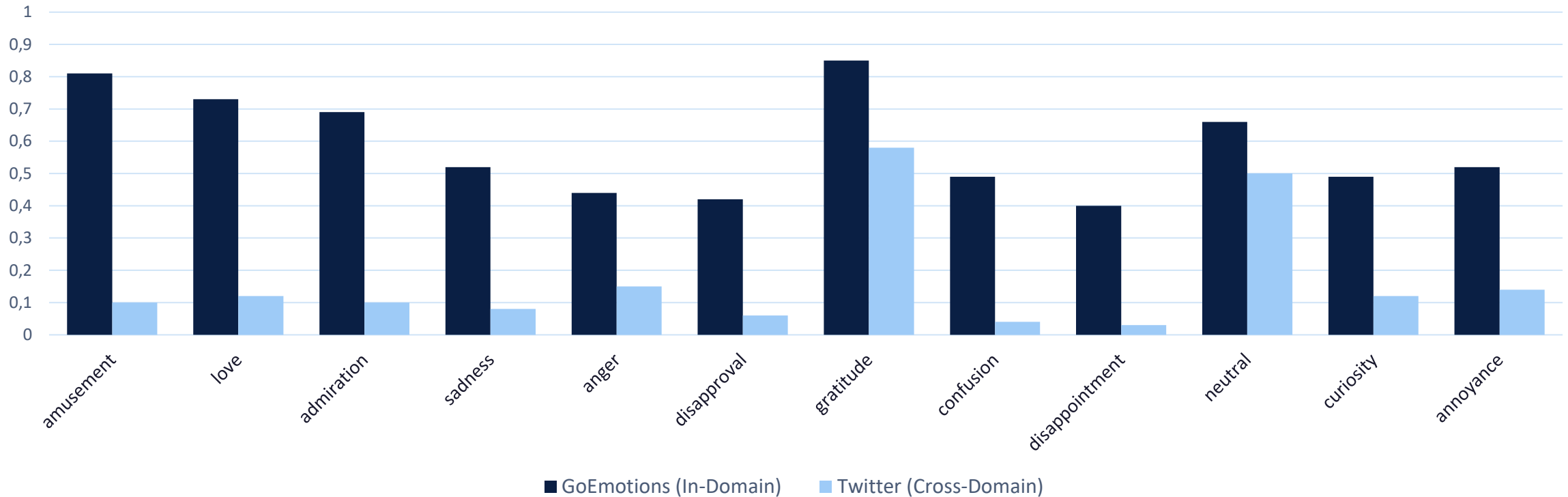
≤10%: Strong transfer

10 to 20%: Moderate

>20%: POOR

Both models show near-identical gaps (~25-26pp). Architecture does not explain the failure. Domain mismatch is the bottleneck.

Per-Emotion Transfer: Where It Breaks Down

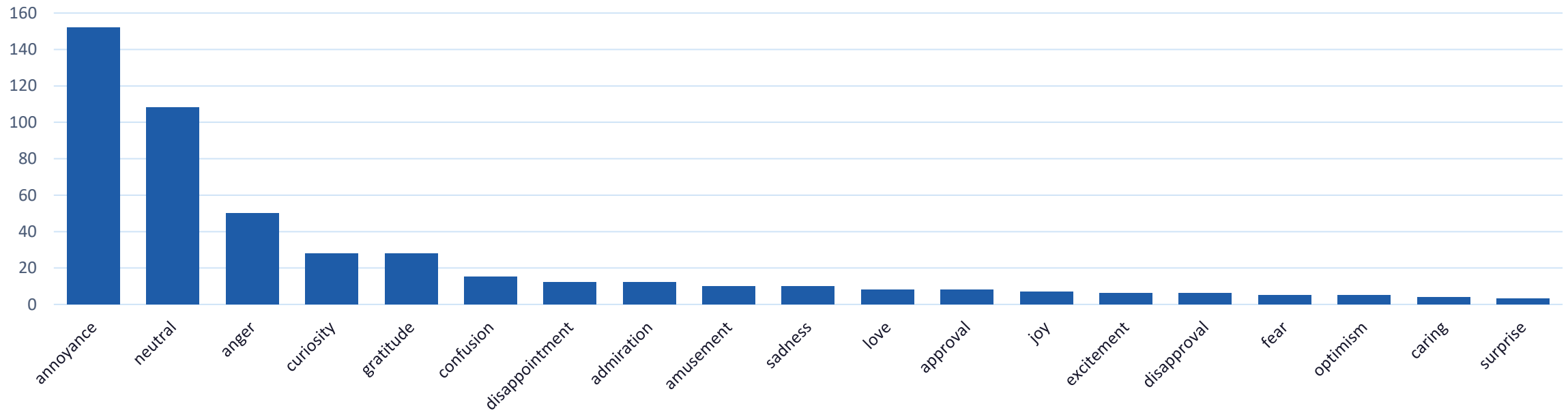


Key Patterns

- ▶ Gratitude transfers best (0.85 to 0.58): "thank you" and "thanks" work the same way across domains
- ▶ Annoyance, the most common customer emotion (30.4%), drops from 0.52 to just 0.14
- ▶ Disappointment nearly vanishes (0.40 to 0.03): customers express it indirectly in support contexts

Emotion Distribution in Customer Support

Count (n=500)



Key Observations

- Annoyance dominates at 30.4% of all annotations, followed by neutral (21.6%) and anger (10.0%)
- Negative emotions collectively account for over 62% of customer support interactions
- The single most important emotion for service routing (annoyance) is also the hardest to transfer

Error Analysis: Where & Why It Fails

313 / 500

tweets misclassified (62.6%) vs 41.5% in-domain error rate

Systematic Failure Modes

Polite Framing

"Thanks, but..." triggers gratitude prediction

Indirect Emotion

Event description without affective words

Lexical Mismatch

Semantic overlap between emotion categories

THE RELIEF ANOMALY

93 annoyance tweets predicted as relief
n=96 training examples, in-domain F1 = 0.00
Model over-activates the worst-learned class under domain uncertainty

Qualitative Examples

"Thanks, but it's been nearly 48 hours..."

True: Annoyance

Predicted: Gratitude

Failure Mode: Polite Framing

"It's going to cost me £42 to return this faulty item."

True: Annoyance

Predicted: Curiosity

Failure Mode: Indirect Emotion

"I can't download any apps, free or otherwise."

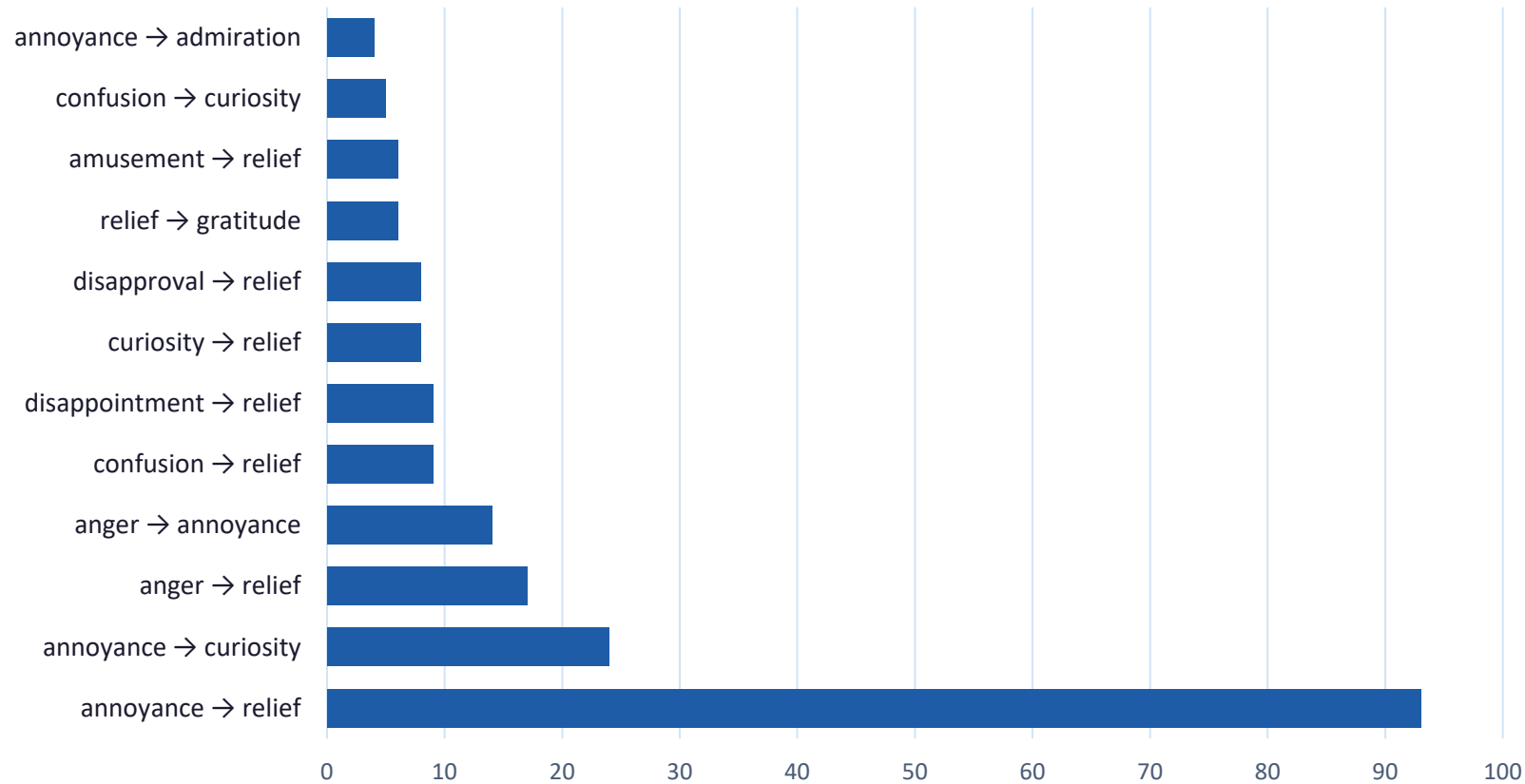
True: Annoyance

Predicted: Neutral

Failure Mode: Event Description

Top Misclassification Pairs on Twitter

Count



Relief Dominance

8 of the top 12 error pairs involve relief as the predicted label.

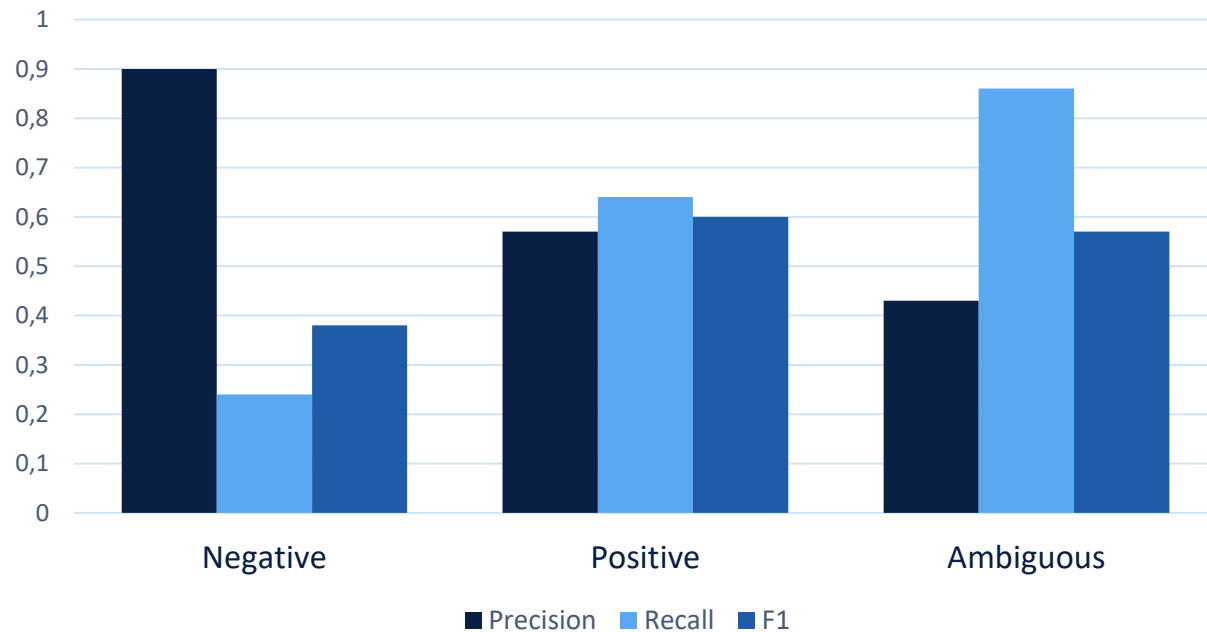
Relief has only 96 training examples and $F1 = 0.00$ in-domain. Under domain uncertainty, the model defaults to this poorly learned class.

Annoyance is the Victim

$93 + 24 = 117$ annoyance tweets misrouted.
That is 77% of all annoyance predictions wrong.

The Grouped Emotion Fix

28 fine-grained classes → 3 groups (Negative / Positive / Ambiguous) Weighted F1: 0.312 → 0.485 (+17.3pp)



Confusion Matrix (True x Predicted)

	Neg.	Pos.	Amb.
Negative	59	18	169
Positive	5	53	25
Ambiguous	2	22	147

The model is nearly always correct when it flags negative (0.90 precision) but misses 76% of truly negative tweets (0.24 recall)

Three Actionable Recommendations

01

Do NOT deploy fine-grained emotion labels without domain adaptation

The relief anomaly shows uncalibrated predictions can actively undermine customer experience, routing frustrated customers away from escalation.

02

Use grouped labels (Negative / Positive / Ambiguous) as a first-pass filter

0.90 precision for negative emotions means confident flagging. Route high-confidence negative messages to senior agents. Annoyance accounts for 30.4% of all customer emotions.

03

Collect 500+ in-domain labelled examples

The annotation process demonstrated that 500 tweets can be labelled with $\kappa=0.733$ following clear guidelines. Domain mismatch, not model size, is the core constraint.

Conclusions & Future Directions

"The answer is: only partially"

- ✓ BERT fine-tuned on GoEmotions: weighted F1 = 0.566 in-domain
- ✗ Transfer to Twitter customer support: weighted F1 = 0.312 (-25.5pp, POOR)
- Architecture does not matter. Domain mismatch is the fundamental bottleneck

Human emotional intelligence remains essential in customer service.

Future Work Roadmap

Domain-Adaptive Pre-Training

Further unsupervised training on unlabelled customer support text (Gururangan et al., 2020). Expected to be most impactful.

Multi-Label Classification

Model co-occurring emotions rather than forcing a single dominant label. Reduces label ordering bias.

Psycholinguistic Feature Augmentation

Add NRC Emotion Lexicon features to the transformer (Zanwar et al., 2022). Improves cross-corpus robustness.